Siddh Patel

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SUMMARY

High school student and hands-on technologist who runs Linux daily, repairs laptops at the component level, and maintains a self-hosted Proxmox homelab. Builds and ships software projects in local AI, reverse engineering, and robotics. LPI Linux Essentials certified, CompTIA A+ in progress, with customer-facing retail experience.

SKILLS

Systems: Linux (Arch, Debian, Ubuntu), Windows, Proxmox virtualization (VMs, LXC, PCIe passthrough), Tailscale, Bash, Git
Programming: Python, JavaScript (Three.js), shell scripting
Hardware & repair: component-level motherboard repair, power-delivery fault isolation, soldering, schematic tracing, multimeter diagnostics, NVMe data recovery, system assembly and bench testing
AI / ML: local LLM serving (Ollama, llama.cpp), QLoRA fine-tuning, gradient-boosted models and ensembling
Security: Windows and Linux hardening, Linux account security (PAM), USB protocol and firmware reverse engineering
Customer & retail: POS and cash handling, end-of-day reconciliation, inventory and returns, clear customer communication

PROJECTS

Byte to Bite: BEST Robotics Simulator — 2026

- Built a 3D browser game recreating the 2026 BEST Robotics field so the team can practice driving and strategy before the real robot exists.
- Added online multiplayer hosted on Google Cloud Run and support for driving with a real VEX EXP controller. Three.js, JavaScript.

Proxmox Homelab and Dual-GPU AI Server — 2023–Present

- Built and maintain a bare-metal Proxmox server running a NAS, AI services, 24/7 AI agents, and test VMs, connected privately over Tailscale.
- Pooled an NVIDIA RTX 3070 Ti and an AMD RX 6700 XT through Vulkan to run 26–30B-parameter language models at 34–51 tokens/sec.

Component-Level Laptop Repair — 2024–Present

- Diagnosed a dead Dell Inspiron motherboard (LA-L241P) to the component level: traced power-delivery schematics to isolate a failed charger IC and restore the board to POST.
- Recovered data directly from the NVMe drive after full board fault isolation.

Reverse Engineering (UGREEN mouse, VEX controller) — 2026

- Decoded the undocumented wireless protocol of a UGREEN mouse receiver and released an open-source Linux settings app for speed, scrolling, and button remaps.
- Worked out how a VEX EXP controller talks to its Brain and exposed it as a system-wide Linux joystick with uinput and Python.

Cybersecurity LLM Fine-Tune — 2026

- Fine-tuned Qwen2.5-7B with a QLoRA adapter on cybersecurity data and served it locally: dataset prep, training, and deployment on my own hardware.

CyberPatriot National Youth Cyber Defense Competition — 2023–Present

- Find and fix vulnerabilities on Windows and Linux systems against the clock, without breaking required services.
- Wrote a read-only Bash diagnostic utility for automated system auditing and documented Linux account-security (PAM) hardening.

WORK EXPERIENCE

Customer Service Associate · Walgreens — May 2026–Present

- Operate the point-of-sale register and handle end-of-day reconciliation accurately in a high-volume store.
- Help customers at the register and on the floor, including device and technical questions, and resolve issues on the spot.
- Coordinate inventory stocking and returns; consistently dependable and on time.

CERTIFICATIONS & EDUCATION

- **LPI Linux Essentials**, certified
- **CompTIA A+** (Core 1 & Core 2), in progress, target Dec 2026
- CompTIA Network+ and AWS Certified Cloud Practitioner, in progress
- **Rock Canyon High School**, expected graduation May 2027
- DCSD Legacy Campus, Secure Network Technologies pathway, currently enrolled